

PEAM



Experience Loop

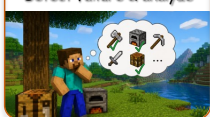
1 Explore

Try sub-tasks



2 Fail & Reflect

Detect failure & analyze



3 Correct & Succeed

Fix and complete



4 Store Trajectory

(failure + correction)



Slow Tier: Deliberative LLM

Reason · Plan · Reflect · Verify
Curriculum · Code · Feedback

Plan & Verify

Execute & Report



Fast Tier: Parametric Skills

Reflexive · Low-latency ·
Execute Consolidated Skills

Episodic Memory

- ✓ Success Trajectories
- ✗ Failure + Correction Pairs
- 📄 Reflection Logs
- 🌐 Environment Observations

Parametric Skills Module

Multimodal MoE-LoRA (Per-Category Isolated Adapters)



Crafting Adapter



Combat Adapter



Gather Adapter

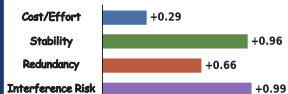


Navigate Adapter

Benefits

- ✓ Parameter-level Continual Learning
- ✓ Forgetting Resistance
- ✓ Fast Inference

1. What to Internalize (What) Parameterization-Worthiness Score

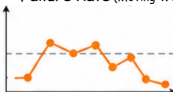


Score ↑ → Worthy!



2. When to Internalize (When) Scale-Free Self-Triggered Consolidation

Failure Rate (Moving Window)



Trigger!

No task-specific failure threshold
Transfers across distributions

3. Parametric Consolidation (How) Isolated Adapters + Failure-Correction Learning



Crafting Adapter



Combat Adapter



Survival Adapter



Physically isolated per category
→ Forgetting resistance by design

The Agent Changes

Skills are internalized
into parameters, not just
stored in a library



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